12.3 General methods of preparation of aldehydes and ketones

A. Preparation of aldehydes and ketones

1.) Oxidation and catalytic dehydrogenation of alcohols

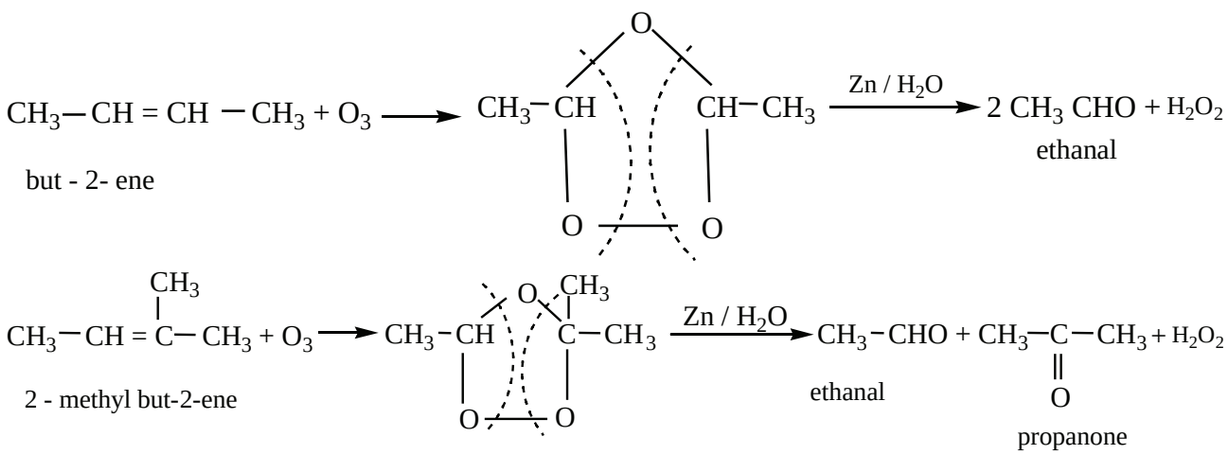
We have already learnt that the oxidation of primary alcohol gives aldehydes and secondary alcohol gives a ketone. Oxidising agents such as acidified $Na_2Cr_2O_7$, $KMnO_4$, PCC are used for oxidation. Oxidation using PCC yield aldehydes . Other oxidising agents further oxidise the aldhydes / ketones in to carboxylic acids (Refer Unit No. - 11 Oxidation of alcohols)

When vapours of alcohols are passed over heavy metal catalyst such as Cu, Ag, alcohols give aldehydes and ketons. (Refer Unit No. - 11 Catalytic dehydrogenation of alcohols

2.) Ozonolysis of alkenes

We have already learnt in XI th standard that the reductive ozonolysis of alkenes gives aldehydes and ketones.

Alkenes react with ozone to form ozonide which on subsequent cleavage with zinc and water gives aldehydes and ketones. Zinc dust removes H_2O_2 formed, which otherwise can oxidise aldehydes / ketones.



Terminal olefines give formaldehyde as one of the product.

Evaluate yourself - 1

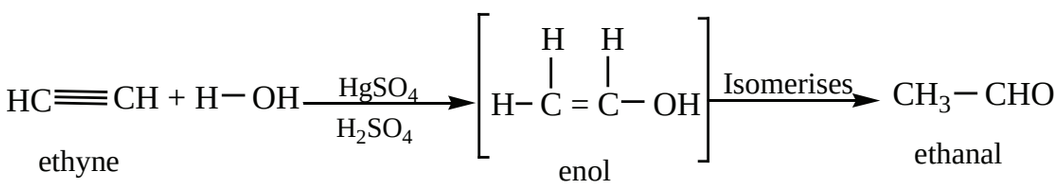
What happens when the following alkenes are subjected to reductive ozonolysis.

1) propene 2) 1 – Butene 3) Isobutylene

3. Hydration of alkynes

We have already learnt in XI standard that the hydration of alkynes in presence of 40% dilute sulpuric acid and 1% $HgSO_4$ to give the corresponding aldehydes / ketones.

a) Hydration of acetylene yields acetaldehyde



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